

Assignment One: Python Basics

Edge computing device programming for AI projects

Student

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Submission file: assignment 1_WangZhihan.py

Deadline: 22/09/2026 11:59pm

GitHub link

<https://github.com/<YOUR-GITHUB-USERNAME>/assignment-one-python-basics>

Description

This is my submission for Assignment 1: Python Basics. All three tasks are implemented in one single Python file, assignment 1_WangZhihan.py, which is stored in the GitHub repository linked above. The program only uses the Python standard library, so it runs on the Jetson Nano without installing anything extra. Starting the file shows a small menu, where each task can be opened on its own or all three can be run one after another.

Task A - Simple Calculator

The calculator asks for the first number, the second number and then the operator (+, -, *, /), and prints the result in a clear form such as "Result: 3 + 5 = 8". A whole expression can also be typed on one line, for example 3 + 5. Input mistakes are handled instead of stopping the program:

- dividing by zero prints "Cannot divide by zero"
- an operator other than + - * / prints "Invalid operation"
- letters typed where a number is expected are asked for again
- a line that is not a proper expression is reported and asked again

Task B - Question Answering Bot

The question is received with input() and the reply is chosen with an if / elif / else chain. The bot supports the five required keywords:

```
hello    -> Bot: Hello! Nice to meet you.
python  -> Bot: Python is a language, and it is very friendly to beginners.
jetson   -> Bot: Jetson Nano is an AI computer made by NVIDIA.
ai       -> Bot: AI means Artificial Intelligence.
name     -> Bot: My name is Python Bot.
```

The matching ignores capital letters and punctuation, so Hello!, PYTHON? and "what is AI" are all understood, and a whole sentence such as "Tell me about the JETSON please" is recognised too. Any other question gets a friendly "Sorry, I don't understand." reply, and typing bye ends the conversation.

Task C - Turtle Drawing

The drawing task uses turtle with for loops, changing colours and changing lengths. It can draw:

- a square, a triangle and a five-pointed star
- a colourful rotating pattern that uses six colours and shrinks each round
- a spider web with twelve spokes in eight different colours

The five pictures can be drawn one at a time or all together. When the drawing is finished, clicking the window returns to the menu.

Testing

The program was tested before submitting. The calculator was checked with all four operators, with decimal numbers, with a division by zero and with wrong input; the bot was checked with the five keywords in upper and lower case and with unknown questions; and the turtle drawing was checked by saving the drawing and confirming that all five pictures are produced. The program did not stop with an error in any of these tests. The checking script, `verify_submission.py`, is included in the repository.

Submitted file: `assignment 1_WangZhihan.py` | Report: `Assignment1_WangZhihan.pdf`

Required file header: `# Name: Wang Zhihan / # Assignment One / # ddl is 22/9/2026 23:59pm`